

CAT® PAVING PRODUCTS

GUIDE TO ASPHALT PAVING

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INTRODUCTION

The Cat Paving Products *Guide to Asphalt Paving* is intended to be a practical reference guide for the process of paving.

This guide deals with asphalt production only to the extent that production of asphalt affects the quality of the material being laid down on a project. Likewise, the design of the various types of bituminous material will be covered briefly in order to help develop an understanding of how mix design may affect the flow of material through a paver and under the paver's screed. The guide is not intended to be a reference for project design or asphalt selection.

The *Guide to Asphalt Paving* will be helpful to all personnel involved with the planning, preparation, and placement of asphalt. Certainly, the crewmembers who operate the paver should understand the information contained in the guide. Crew supervisors who help the operators and work on the planning steps will find the guide useful. Estimators, project managers and project superintendents can use the guide to help plan project efficiency and prepare to meet the requirements and specifications of various applications. Finally, quality control and quality assurance personnel will find valuable troubleshooting information in the guide.

Specific applications will be discussed in detail. However, due to the many variables in asphalt paving around the world, it is impossible to cover every situation in a reference guide such as this. Also, the terms used to describe paver components, asphalt mixes, and project variables are different around the world. When possible, for the purpose of clarity, terms will be explained or defined in different ways. For example, in this guide, the word "asphalt" will be used most often to describe what may be called "bituminous material" or "asphalt concrete" in some parts of the world.

No matter how the material being laid down by the paver is referred to, there are four basic elements to quality construction of asphalt layers. The information contained in this guide will consistently refer back to these four elements.

BASICS OF PAVING QUALITY

1. **Utilize fundamentals correctly every time.** When the crew violates certain asphalt paving fundamentals, defects in the asphalt layer usually appear.

   A crew that is well-trained in paving fundamentals and takes the time to control the fundamental aspects of paving will generally be successful. In Unit 3, "Fundamentals of Paving," these basic principles are described. They should be the foundation for all crew actions. In Unit 8, "Troubleshooting Guide," the guide demonstrates that adherence to these fundamental principles creates solutions to many common mat defects.



A well-trained crew has set up the feeder system to maintain the correct head of material.

2. **Pave efficiently.** To pave efficiently means to calculate and maintain a paving speed that consumes the mix in a steady manner and avoids long paver stops waiting for mix.

   A calculated paving speed should be used on any project that has long, uninterrupted paving segments such as a highway. Minimizing or eliminating paver stops helps to maintain smoothness and the consistency of layer temperature in front of the compaction process. Paving continuously for long periods on projects such as parking lots and city streets, is much more difficult, but striving for efficiency is still worthwhile. How to calculate paving speeds and plan for efficiency are covered in Unit 2, "Pre-project Planning."



Efficient paving means continuous paving on highway projects.

3. **Understand grade and slope requirements.** On many projects, automatic grade control and/or automatic slope control will be required to create the correct layer thickness (yield), improve smoothness, match the height of an existing structure, and/or to create profile (slope).

   Before starting to work, the project's requirements must be determined and grade and slope control must be set up to make the paver create the desired results. A paver is not set up to pave a parking lot the same way it is set up to pave an airport runway or a highway.

   Each project has different requirements and specifications. In Unit 6, "Automatic Grade and Slope Control," the basics of grade and slope control, as well as specific tips for controlling yield, smoothness, joint matching, and so forth are discussed.



Set up grade and slope control according to project requirements or specifications.

4. **Avoid big mistakes.** Big mistakes, like spills in front of the paver or running low on material in the auger chamber, cause big quality problems and often result in costly rework.

   Throughout this guide, examples of paving mistakes will be shown with suggestions for learning how to avoid them. Many times, lack of planning and poor preparation contribute to mistakes. The equipment may be used incorrectly. Or, the crew may be skipping set-up steps and violating paving fundamentals.

   Whatever the cause, big mistakes should be easy to identify and correct.



Spills on the grade in front of the paver are a big mistake and should be corrected.


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Unit 1
THE BASICS OF ASPHALT PRODUCTION

It is essential that all crewmembers know the basics of asphalt production and how different mix designs affect the paving process. Effective communications between the plant personnel and the paving crew are crucial, as the paving crew must be aware of changes in hourly production or mix formulation.

[BASICS OF ASPHALT PRODUCTION]

In general terms, asphalt is a combination of aggregates, fine material, and bitumen, also called asphalt cement. The aggregates, when moved into close contact during the compaction process, provide strength in the asphalt layer. The bitumen and fines fill the voids between the aggregates and act as the binder that helps bond the aggregates.



Depending on the specifications mandated by the project owner or by the desired end result, additives and modifiers are often included in the mix formula. Various types of recycled material, most commonly reclaimed asphalt pavement, may be included in the mix formula. In Unit 1, the basics of asphalt production and the types of asphalt mixes and their uses are covered.



[ASPHALT MANUFACTURING FACILITY]

An asphalt manufacturing facility consists of many elements: aggregate stockpiles, asphalt cement storage tanks, an aggregate feeder system, an aggregate drying system, a system to blend aggregate and asphalt cement, a production control system, dust collectors, surge and storage bins (silos), and a scale. The facility will often have a laboratory used for quality control and quality assurance. A location where trucks are cleaned out and a suitable release agent is applied to the bodies of the haul units is typical.

[AGGREGATE STOCKPILES]

Stockpiles usually consist of crushed aggregate in different sizes and sand (fine material), although there may also be stockpiles of what is commonly called river run, or uncrushed aggregate. Each stockpile should be clearly marked so the loader operator who transports material can easily identify aggregate type and size. Production of asphalt should not begin unless there is at least one day's supply of aggregate for each type scheduled to be used in the manufacturing process.

Stockpiles should be located in such a manner as to prevent contamination through contact with adjacent stockpiles. Barriers are commonly used to separate stockpiles.

Placing aggregates on paved surfaces is another way to minimize contamination of stockpiled material. The loader operator will not have to worry as much about introducing soil or other low-quality materials into the asphalt manufacturing process.

Reducing the moisture content of stockpiled material is another issue. When aggregates have high moisture content, additional drying time will be required, which is costly due to increased fuel consumption by the plant and a decrease in hourly production.

Protecting stockpiles with removable covers is a common practice. Another quality control step that is becoming increasingly popular is placement of stockpiles inside covered structures. The additional costs of these techniques can be justified when production costs go down and hourly production increases.



A well organized asphalt plant increases productivity and profitability.



Depending on the design of the asphalt manufacturing facility, stockpiles are usually created in three ways: with stationary conveyors, moving conveyors or dump trucks.

If the plant is located within a quarry and crushing operation, aggregates are stockpiled by discharging crushed material from stationary belt conveyors or from moving radial belt conveyors. The stationary conveyor creates a cone-shaped stockpile. The moving radial conveyor creates a fan-shaped stockpile. When aggregates are transported to a dedicated asphalt manufacturing facility, dump trucks are commonly used.

When discharging material from a conveyor or from a dump truck, material segregation may occur—even though the material is theoretically of a uniform size. The fine (smaller) material tends to flow toward the center of the stockpile, while larger material rolls toward the outer edges of the stockpile.

The loader operator who is transferring material from the stockpiles to the aggregate feeder system (cold feed bins) can help blend material. The loader operator should not take material from the same stockpile location in successive passes.

Instead, the loader operator should divide the face of the stockpile into three segments. Take each bucket from a different segment in an alternating sequence. Approach the stockpile with the loader bucket lowered to the ground, but not digging into the soil.

If the stockpile is not covered and there is moisture in the lower portion of the stockpile, raise the bucket 0.5 m to 1.0 m (18" to 36") off the ground when approaching. Slowly curl the bucket upward and back as it enters the stockpile. When dumping each load, slowly rotate the bucket from the lowest height possible and fill the appropriate feeder bin.

Visual observation of the stockpile will identify areas of segregation, if any exist. Samples of the mix produced at the plant will also help determine if the gradation of the mix is within specifications.

[ASPHALT CEMENT STORAGE TANKS]

Plants typically include multiple storage tanks that contain the various types of asphalt cement used in the production of different mix designs. Each tank should be clearly labeled, indicating the type of asphalt cement that is stored and the capacity of the tank. Tank piping used for loading and unloading should be clearly labeled as well.

Environmental regulations vary around the world, and permits for above-ground and in-ground asphalt cement storage tanks are frequently required. The permits spell out the maintenance requirements and necessary steps to prevent leaks and spills in the storage area.

Asphalt cement storage tanks are insulated and include heating systems and often mixing systems, especially if the asphalt cement has been modified with polymers. For example, unmodified asphalt cement may be heated to 160° C (320° F) in the storage tank. Polymer-modified asphalt cement will be heated and stored at slightly higher temperatures.



Storage tanks for asphalt cement.

The asphalt cement storage tank supplies the mixer with the specified amount of asphalt cement.

[AGGREGATE FEED SYSTEM]

A feed system supplies the mixer with aggregates of the correct size and in the correct amounts according to the gradation specified by the mix design.

The facility must have a sufficient number of feeder bins to supply the various mix elements. One feeder bin should not be used to supply two types or sizes of aggregates.

Feeder bins are normally open-top compartments with sloped sides and adjustable gate openings at the bottom for controlled discharge of material. It is the responsibility of the loader operator(s) to maintain adequate levels in the feeder bins.

During the mix manufacturing process, the feeder bins should not be allowed to run extremely low or completely out of material. Most control systems have a bin level detection system that automatically stops the flow of material to the dryer and mixer if one or more bins run out of material.



The feed system is designed to deliver a steady flow of aggregate to the mixer.

Material sizing screens are an integral part of the aggregate feeder system. The sizing screens prevent oversize aggregate from entering the dryer and mixer. The sizing screens have one-dimension openings. For example, if a base mixture is being manufactured, the screen openings may be 45 mm (1.75"). For the production of binder and surface mixes, screens will usually have an opening that is up to 6 mm (0.25") larger than the largest nominal aggregate size.

User Tip: *Oversize material that shows up in the asphalt layer is a common problem. There can be several causes. One is the asphalt manufacturing facility using the wrong size opening in the sizing screen. Or, it may be that the sizing screen has been damaged and has holes. When troubleshooting the appearance of oversize material in the asphalt layer, include an inspection of the sizing screens in the checklist.*

Aggregates flow out the discharge gates in a metered fashion, usually onto a moving conveyor belt. Scales, or load cells, weigh the contents of the belt. This information is sent to the facility's control system which regulates the speed of the conveyor belt according to the production needs of the dryer and mixer.

[DRUM MIXER]

Various types and sizes of drum mixers are available. Their operation can be summarized in the following manner: dry and heat the aggregates, introduce asphalt cement, blend the aggregates and asphalt cement, discharge the mixture, and provide an exhaust for gases and dust. These steps produce a continuous flow of mix into surge bins (silos) or storage bins.

Drum mixers are the most recent type of drying and mixing equipment. They can be permanent or portable. Hourly production rates are determined by drum length and diameter.



A drum mixer's spiral flighting continuously moves aggregate to the lower end.



Plants that utilize a drum mixing system are capable of high-output, continuous asphalt production.

The mixer's production rate is affected by the moisture content and the desired output temperature. Aggregate having high moisture content requires longer drying time resulting in lower hourly production. Likewise, certain mixes require higher output temperatures that vary according to the type of bitumen, the length of haul to the project, ambient conditions, and the compaction process.

User Tip: *It is important for the crew foreman or project superintendent to receive accurate hourly production rates. Hourly production is one determining factor for ordering trucks to haul mix to the project. Knowing hourly production also helps the supervisors and crewmembers calculate the working speed that promotes efficient paving. Remember, paving efficiently is a key quality element.*

Drum mixers are installed at a slight inclination. Aggregates enter the upper end of the drum mixer. A burner blower also extends into the upper end of the drum mixer. The burner blower provides the heat that dries the aggregates and brings their temperature to a specified level.

Spiral flighting inside the drum mixer continuously moves the aggregates toward the lower end of the drum mixer. It usually rotates at a rate of five to 10 revolutions per minute.

The line that discharges liquid, asphalt cement extends into the lower end of the drum mixer. As the aggregates move from the middle area of the drum mixer toward the lower end (discharge), they are coated with hot asphalt cement.

Some gases and dust are created inside the drum mixer during the drying and mixing process. An exhaust fan pulls the gases and dust out of the drum through a vent. The suction created by the exhaust fan also helps air induction at the blower burner end of the drum, leading to better combustion.

Hot mix flows out the bottom of the drum mixer's lower end, usually onto a drag slat conveyor that loads the mix into a surge or storage bin (silo).



[BATCH MANUFACTURING]

A batch manufacturing facility produces one quantity of mix at a time. A batch facility has many of the same components as a drum mix system. The aggregate stockpiles and feeder system operate in much the same manner as described in the section on drum mixers. The main difference between drum mixers and batch facilities is how the aggregates are heated and blended with asphalt cement.

In a batch facility, the various cold aggregates are delivered by a belt system to a rotary dryer. The aggregates enter the upper end of the rotary dryer, while the burner blower is at the lower end. Spiral flighting and gravity move the aggregates through the flame. The aggregates and hot gases inside the dryer move to the lower end. The hot aggregates then are discharged into an elevating system.

The bucket elevator, an enclosed container, transfers the hot aggregates to the top of the batch tower. While the aggregate proportions have been initially controlled by the cold feeder system, a more precise screening occurs at the top of the batch tower. A series of vibrating screens separates the aggregates by size and fills the hot aggregate bins. From the hot aggregate bins, the aggregates are transferred in measured amounts to the mixer, usually a twin-shaft pugmill.

User Tip: *The vibrating screens typically handle only a small amount of material. If a screen is overloaded, material that would normally pass through a given screen opening may run over into an adjacent hot bin. The gradation of the aggregates in the mix would, therefore, be incorrect. When troubleshooting mix conformance problems, it is important to sample the hot bins to assure that overloaded screens are not causing the problem.*

When the correct amounts of aggregates are transferred to the mixer, the container is closed. Hot aggregates are mixed for a specified time. A metered amount of hot asphalt cement is injected into the pugmill and the wet mixing cycle begins. It is important that the volume of material inside the mixing pugmill be maintained at the correct level to assure correct blending and aggregate coating.

At the end of the mixing cycle, the "batch" can be loaded directly into a haul unit, but is more commonly retained in a surge or storage bin (silo) for later loading.

[DUST COLLECTORS]

Dust collection systems are an important part of air quality control. There are two types of dust collection systems: a wet collector in the form of a high-pressure venturi or a fabric filter collector, commonly called a baghouse.

Wet collectors are less common because the collected dust is not available for use as a component in approved mix designs. Also, the water and dust mixture is a waste product that requires special handling and disposal.

Dry dust collectors force the dusty exhaust air produced during the manufacturing process (drum mixer or batch mixer) through a series of cloth bags. Dust clings to the outside of the bags and clean air exits from the baghouse through a filtered air outlet. The rows of bags receive periodic cleaning by means of air pulses and vibration. One row of bags is cleaned at a time. The dust falls into a hopper at the bottom of the baghouse.

To prevent contamination, many dust collection systems have an automatic shutoff feature that stops production when the collection hopper is full.



Dry dust collected by a baghouse has the potential to be returned to the mix.

Baghouse dust may be returned to the drum mixer or batch mixer, if the mix design permits inclusion of baghouse dust.

User Tip: *If clumps of dust appear in the mix, it may be a sign that bag cleaning is not being done properly. Be alert for dust balls or "fat spots," if the mix contains baghouse dust.*

[SURGE OR STORAGE BINS (SILOS)]

Surge or storage bins hold the material produced by the asphalt manufacturing facility until the mix is loaded into haul units and transported to the worksite.

By definition, a surge bin is intended as a temporary holding unit. The surge bin allows the production of mix to continue, even if there is an interruption in the availability of haul units. A batch facility is capable of suspending operation while waiting for haul units, if there is no surge bin. A drum mixer is designed for continuous operation and does not produce high quality material if the drum mixer stops and starts.



If a surge bin is used in conjunction with a batch facility, production of mix can begin prior to the scheduled arrival of haul units. Hauls units can quickly load from a surge bin without the need to wait for the completion of a batch cycle. Therefore, the cycle times of haul units are more predictable, and the task of ordering haul units should be less complicated because of the continuous flow.

Depending on the capacity of the surge bin, the batch mixer or drum mixer can cease production when there is enough surge capacity to fill the haul units that are scheduled for the remainder of the shift. The cost of a surge bin is repaid quickly by the reduction in facility operating costs and may reduce the number of required haul units.



Storage bins provide additional capacity for the manufacturing facility. An asphalt manufacturing facility may have multiple storage bins that permit the mixer to produce and store different types of mixes. The facility can simultaneously serve different customers and different projects.

Storage bins are heated and insulated. Mix may be stored for the duration of one paving shift in some storage bins. Other long-term storage silos can hold mix up to three days at 160°C (320° F).

The special silos have one outlet flap, which is sealed with oil to prevent air from entering through the flap and passing upward through the stored mix. Storage bins promote maximum production at the manufacturing facility because they can hold many hours of mixer output. The increased manufacturing production means increased paving production. Consider the following examples.

A drum mix asphalt plant with a surge bin and no storage bins is going to produce 200 tonnes (220 tons) per hour. A 3 m (10') paver with vibratory screed being supplied by a material transfer device is laying down the mix at an uncompacted depth of 65 mm (2.5") and a constant width of 5.5 m (18'). At a paving speed of 4.5 m per minute (14' per minute), the paver will consume the hourly mix production in an efficient manner. In 10 hours, for example, the paving crew will lay down 2 000 tonnes (2,200 tons). The paver, which theoretically can lay down 1 000 tonnes (1,100 tons) per hour, actually has unused capacity at this production rate.

Now, let's add 300 tonnes (330 tons) of storage capacity to the manufacturing facility. Mix production begins two hours before the arrival of haul units and the storage bin is filled before the start of paving.

Since the crew will work 10 hours, divide the storage capacity by 10. Add the result—30 tonnes (33 tons)—to the hourly production rate. Hourly paving production will be 230 tonnes (250 tons) per hour. Paving speed is increased to 5 m per minute (16' per minute) to consume the additional mix in an efficient manner.

At the plant, the facility operator loads trucks primarily from the surge bin to keep the mixer working continuously, and periodically from the storage bin in order to use the storage capacity by the end of the shift. Daily production is now 2 300 tonnes (2,500 tons) in the same 10-hour period. One or two additional haul units will be required.

User Tip: *While surge and storage bins have various designs, most have clam-type discharge gates and cones that are intended to minimize material segregation during truck loading and discharge. If any of the gates are malfunctioning, segregated material may appear in the haul units and in the asphalt layer. Likewise, if the level of mix in a bin goes below the top of the cone at the bottom of the bin, segregation may occur. Include bin inspection in the list of troubleshooting steps when trying to determine the cause of segregation.*

[SCALES]

The weight of the mix loaded into haul units at the manufacturing facility must be carefully controlled and documented. Haul units must not carry loads that exceed the rating of the haul unit or the maximum axle load specified by local public works departments.

Load-cell scales are common under or on the sides of surge and storage bins. They measure the empty and loaded weights of the haul unit. A computerized system calculates net load weight.

Some surge and storage bins measure the weight of the mix in the bin before and after discharge in order to control and document the loads carried by haul units.

The driver of the haul unit receives a load ticket that is delivered to a collector at the worksite. Usually the project owner receives a copy of each weight ticket and the contractor also keeps a copy. Weight tickets are the basis for payment on many projects. Analysis of weight tickets is helpful for calculating yield accuracy and for verifying a paving crew's hourly production rate.



Surge and storage bins help assure uninterrupted flow of material to the worksite.

User Tip: *Loading time is stamped on the weight ticket. Write the arrival time at the worksite on the ticket. Knowing the time it takes for the haul units to make the trip from the manufacturing facility to the worksite is useful when ordering haul units. This may help more accurately adjust the number of haul units throughout the shift to account for rush hour traffic and normal traffic flow in urban areas.*

[MANUFACTURING FACILITY CONTROLS]

Modern manufacturing facilities are controlled by complete computer systems with varying degrees of operational complexity. The control systems are housed in portable or stationary control houses.

Normally, control houses are positioned to provide the operator with a good view of the facility's operations. The control house contains the computer controls and a wraparound console with various computer screens. The facility operator can simultaneously monitor mix production, mix storage and truck scale operation.



The control system stores the data for various mix formulas and receives input from a variety of measuring devices, such as burner temperature sensors, aggregate feeder scales, and mix output temperature sensors. The system automatically adjusts conveyor belt speeds, burner temperature, asphalt cement flow rates and mixing times.

From the standpoint of the paving crew, plant operation has one primary goal—produce uniform, high-quality mix at the estimated hourly production rate. The operator of the manufacturing facility must have good communication with the paving crew. When something changes in the mix manufacturing process, the paving crew has to know what that change is and how it will affect the arrival of the material at the worksite.

[BITUMEN]

What is technically known as bitumen may also be called asphalt cement or asphalt binder in many parts of the world. Naturally occurring bitumens were the original binders used in the earliest production of bituminous mixtures used for paving. The supply of natural bitumens is limited and only accessible in a few locations. As the demand for bituminous paving material increased in the first half of the 20th century, other sources of bitumen had to be developed.

Today, the bitumen used in the production of paving material is produced by the distillation of crude oil during the refining process. At ambient temperature, bitumen is a semi-solid. Bitumens soften and eventually become liquid when sufficiently heated.

Bitumens are produced and classified based on their ability to satisfy project engineering requirements. Until the 1990s, bitumen specifications in Europe and North America relied primarily on mechanical tests for hardness and viscosity.



Heated bitumens added to the mix provide the required binder.